*****SCHEDULED DOWNTIME NOTICE*****...

Status	Finished
Started	Tuesday, 3 March 2026, 8:10 AM
Completed	Tuesday, 3 March 2026, 8:50 AM
Duration	39 mins 55 secs
Grade	**6.67** out of 18.00 (**37.04**%)

Question 1

Correct

Mark 2.00 out of 2.00

[2 marks / 2 *punte*]

A UART port is configured to operate at 57600 baud, with 8-bit data, one start bit and one stop bit, and even parity bit. What is the maximum data transfer rate on the channel, in bytes per second (rounded to a full byte)?

'n UART poort word opgestel vir dataoordrag teen 57600 baud, met 8 databisse, een begin- en een stop-bis en ewe pariteitsbis. Wat is die maksimum dataoordragtempo op die kanaal, in heel grepe per sekonde (gerond tot 'n vol greep)?

Answer: 5236

Question **2**

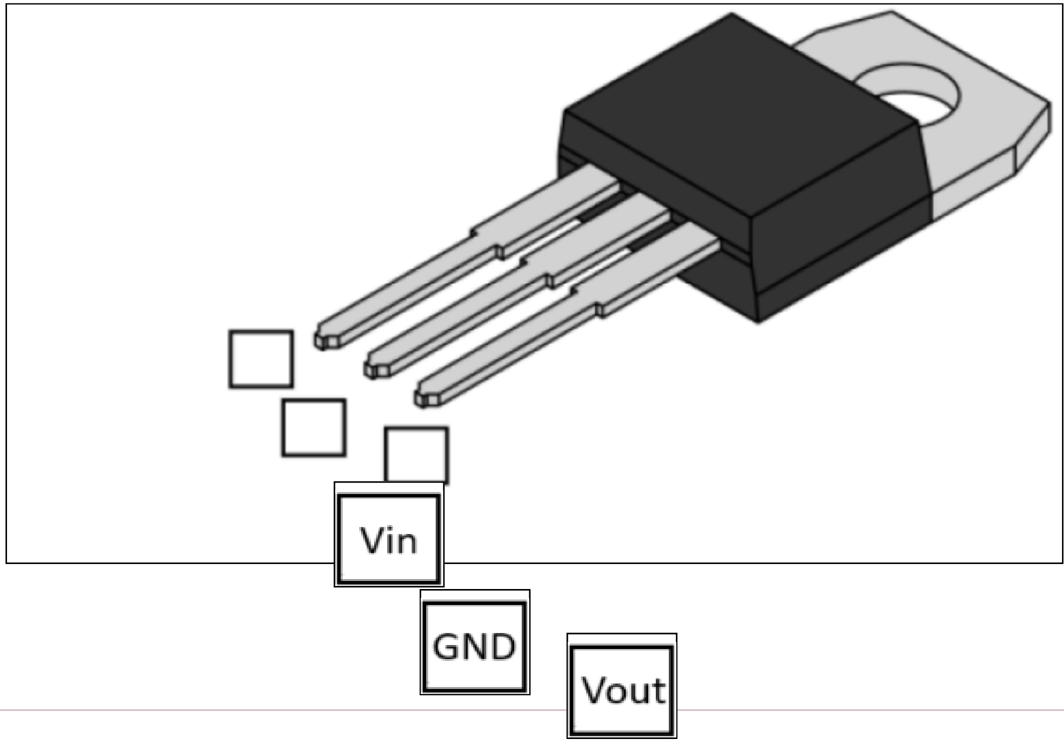
Incorrect

Mark 0.00 out of 1.00

[1 mark(s) / *1 punt(e)*]

Using the LD1117 datasheet place the pin names in the correct places on the image.

Gebruik die LD1117-datablad en plaas die penname op die regte plekke op die prent.



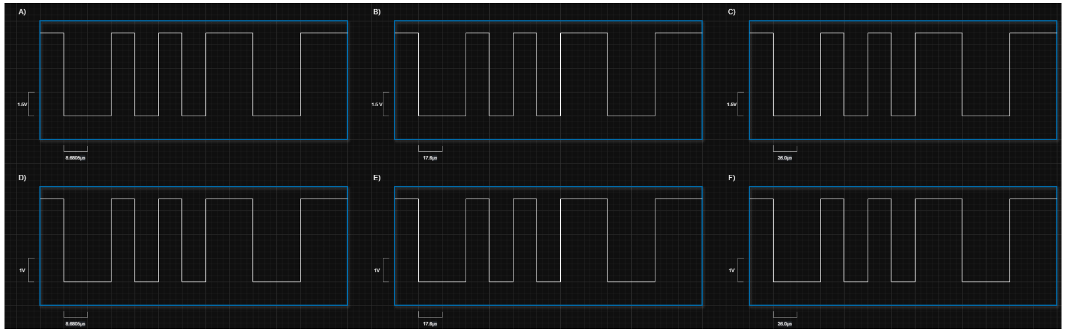
Question 3

Incorrect

Mark 0.00 out of 3.00

[3 marks / 3 *punte*]

Consider the following set of oscilloscope readings from measuring sent UART data: (Open the image in a new tab to zoom in) *// Oorweeg die volgende stel ossilloskooplesings vanaf die meting van gestuurde UART-data: (Maak die prent in 'n nuwe oortjie oop om in te zoem)*



A) Which of these measurements above would you expect to be sent from your Design (E) - 314 project device, if you instantiated the UART settings properly? *// Watter van hierdie metings sou jy verwag om vanaf jou Ontwerp (E) - 314-projektoestel gestuur te word as jy die UART-instellings behoorlik geïnstantieer het?*

D

B) Consider now the following UART instantiation variables from the STM32Cube code of another project. *// Oorweeg nou die volgende UART-instansiasieveranderlikes van die STM32Cube-kode van 'n ander projek.*

```
huart2.Instance = USART2;
huart2.Init.BaudRate = 115200;
huart2.Init.WordLength = UART_WORDLENGTH_9B;
huart2.Init.StopBits = UART_STOPBITS_1;
huart2.Init.Parity =  UART_PARITY_EVEN;
huart2.Init.Mode = UART_MODE_TX_RX;
huart2.Init.HwFlowCtl = UART_HWCONTROL_NONE;
huart2.Init.OverSampling = UART_OVERSAMPLING_16;
huart2.Init.OneBitSampling = UART_ONE_BIT_SAMPLE_DISABLE;
huart2.Init.ClockPrescaler = UART_PRESCALER_DIV1;
huart2.AdvancedInit.AdvFeatureInit = UART_ADVFEATURE_NO_INIT;
```

Which measurement would you expect to be sent from this project device? *// Watter meting sou jy verwag om vanaf hierdie projektoestel gestuur te word?*

E

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	DEL

Source: www.LookupTables.com

What is the ASCII character being transmitted via UART in all the readings? // *Wat is die ASCII-karakter wat in al die lesings via UART oorgedra word?*

A8

Question 4

Partially correct

Mark 1.00 out of 2.00

[2 Marks // 2 Punte]

Match the given Git command to it's corresponding description from the provided options. *// Pas die gegewe Git-opdrag by die ooreenstemmende beskrywing uit die verskafte opsies.*

git add [file]	add a file as it looks now to your next commit (stage).// voeg 'n lêer soos dit nou lyk by jou volgende commit (stadium).
git status	show the commit history for the currently active branch.//wys die toewydingsgeskiedenis vir die tans aktiewe tak
git clone [url]	retrieve an entire repository from a hosted location via URL. // haal 'n volledige bewaarplek vanaf 'n gehoste ligging via URL op.
git branch	show the commit history for the currently active branch.//wys die toewydingsgeskiedenis vir die tans aktiewe tak

Question 5

Incorrect

Mark 0.00 out of 2.00

[2 Marks / 2 Punte]

For your D2 LED used in the project, what is the primary design current limit to still have 3.3V across the LED and resistor combination?

Vir jou D2 LED wat in die projek gebruik word, wat is die primêre ontwerpstroomlimiet om steeds 3.3V oor die LED en resistor kombinasie te hê?

Select one:

- ○ a. GPIO absolute maximum current as stated in the STM32 datasheet / *GPIO absolute maksimum stroom soos vermeld in die STM32 datablad*
- ○ b. The usable brightness level of the LED as observed by the designer / *Die bruikbare helderheidsvlak van die LED soos waargeneem deur die ontwerper*
- ◉ c. LED operational current as stated in its datasheet / *LED operasionele stroom soos vermeld in sy datablad* ⊗
- ○ d. LED absolute maximum current as stated in its datasheet / *LED absolute maksimum stroom soos aangedui in sy datablad*
- ○ e. GPIO operational current as stated in the STM32 datasheet / *GPIO operasionele stroom soos vermeld in die STM32 datablad*

Question **6**

Partially correct

Mark 1.00 out of 2.00

[2 marks / 2 punte]

You tested your project on your bench using 9V from the bench supply and the USB cable connected to a terminal program set to 8E1 and 57600 BAUD. You got the correct system responses during these tests. When you demo your board, there is no communication between your device and the teststation. Mark all of the following options that could cause the problem.

Jy het jou projek op jou bank getoets met 9V vanaf die banktoevoer en die USB-kabel gekoppel aan 'n terminaalprogram wat op 8E1 en 57600 BAUD gestel is. Jy het die korrekte antwoorde van jou stelsel tydens hierdie toetse gekry. Wanneer jy jou bord demonstreer, is daar geen kommunikasie tussen jou toestel en die toetsstasie nie. Merk al die volgende opsies wat die probleem kan veroorsaak.

Select one or more:

- [ ] a. Your STM32 Jumper is set to U5V and not E5V / *Jou STM32-jumper is op U5V gestel en nie E5V nie*
- [ ] b. The UART parity bit is set incorrectly / *Die UART-pariteitbis is verkeerd gestel*
- [ ] c. The 3.3V regulator is soldered in the wrong way round / *Die 3.3V-reguleerder is verkeerd gesoldeer*
- [ ] d. The HAL_UART_Transmit() function is never called / *Die HAL_UART_Transmit() funksie word nooit genoem nie*
- [x] e. The RX and TX lines are not making a good connection / *Die RX- en TX-lyne maak nie 'n goeie verbinding nie* ✓
- [ ] f. The RX and TX lines are not wired to the TIC / *Die RX- en TX-lyne is nie aan die TIC bedraad nie*
- [x] g. The UART TX and RX lines are swapped / *Die UART TX- en RX-lyne word omgeruil* ✓
- [ ] h. The UART BAUD rate is set incorrectly / *Die UART BAUD-koers is verkeerd gestel*
- [ ] i. The jumper J30 is not soldered in / *Die konneksie J30 is nie ingesoldeer nie*
- [ ] j. The HAL_UART_ReceiveIT() function is never called / *Die HAL_UART_ReceiveIT() funksie word nooit genoem nie*

Question 7

Incorrect

Mark 0.00 out of 2.00

An STM32 MCU runs at 84 MHz.
An LMT01 sensor outputs pulses at 88 kHz.
The LMT01 is connected to a GPIO pin as an external interrupt on the STM32 (rising edge triggered)

How many full MCU clock cycles occur during one LMT01 sensor pulse period?
//
'n STM32 MCU loop teen 84 MHz. 'n LMT01-sensor gee pulse teen 88 kHz uit. Die LMT01 is aan 'n GPIO-pen gekoppel as 'n eksterne onderbreking op die STM32 (stygende rand geaktiveer). Hoeveel volle MCU-kloksiklusse vind plaas gedurende een LMT01-sensorpulsperiode?

Answer: 954.5 cycles per pulse

Question 8

Partially correct

Mark 2.67 out of 4.00

[4 marks / 4 *punte*]

Below is an attempt at writing code by a first-year student you are tutoring. They've asked you to help them fix it since they are utterly lost when it comes to interpreting the error messages. Change the code such that it compiles, is logically correct, and performs the intended functions. DO NOT make any changes to the function headers as they are crucial to their codebase.

(No penalty applied for multiple checks.)

//

Hieronder is 'n poging om kode te skryf deur 'n eerstejaarstudent wat u onderrig. Hulle het jou gevra om hulle te help om dit reg te stel, aangesien hulle heeltemal verlore is wanneer dit kom by die interpretasie van die foutboodskappe. Verander die kode sodat dit saamstel, dit logies korrek is en die beoogde funksies verrig. MOENIE enige veranderinge aan die funksieopskrifte aanbring nie, aangesien dit deurslaggewend is vir hul kodebasis.

(Geen penalisering vir veelvuldige toetse.)

Answer: (penalty regime: 0 %)

Reset answer

```
uint16_t collatz(uint16_t ColConj)
{
    //A function meant to perform one step of the Collatz Conject
    //If the input number is even, it halves it.
    //If the input number is odd, it multiples it by 3 and adds 1
    if (ColConj % 2 == 0)
    {
        ColConj = ColConj / 2;
    }
    else
    ColConj = ColConj * 3 + 1;
    return ColConj;
}

uint16_t bwiseOR(uint16_t bwise1, uint16_t bwise2)
{
    //Perform a bitwise-OR operation on the two input numbers and
    return(bwise1||bwise2);
}

uint16_t ConvertRatio(uint16_t ConvInput)
{
    //Convert the given value, which ranges from 0 to 512, to a n
```

```c
    //While maintaining the same ratio.
    //ie. 256 -> 500
    ConvInput = (ConvInput/512)*1000;
    return ConvInput;
}

uint16_t Weirdifyer(uint16_t input, uint16_t mode)
{
    //Manipulates an input number based on the mode
    //if the mode is 0, it simply returns the number
    //if the mode is 1, it multiplies the number by 3
    //if the mode is 2, it adds 1 to the number
    //otherwise it sets the number to 0
    if (mode == 0)
            input = input;
    else if (mode == 1)
            input = input*3;
    else if (mode == 2)
            input = input+1;
    else
    input = 0;
    return input;
}
```

	Test	Input	Expected	Got	
✓	Collatz	0	ok	ok	✓
✗	Bitwise Or	1	ok	expected 79, got 1	✗
✗	Convert Ratio	2	ok	expected 19, got 0	✗
✓	Weirdifier	3	ok	ok	✓
✓	Syntax correct	4	ok	ok	✓

Show differences

Partially correct

Marks for this submission: 2.67/4.00.